Seonghun **Bae**

ROBOTICS & FUTURE MOBILITY LEARNER

Nam-gu, Ulsan, Republic of Korea

(+82) 10-9598-0665 | baeseonghun@naver.com | https://blog.naver.com/seonghun-s | seonghun00

"Fail forward, grow upward."

Summary

This is Seonghun Bae, who wants to become robotics engineer seeking to specialize in autonomous navigation and mobile robot (AMR) control systems. My main interests are in ROS2, SLAM, Navigation, simulation and mobile robot control systems. I always try to be a humble learner, listening to feedback from senior engineers and growing together with my colleagues. To improve my skills every day, I carefully record and share my project journeys on GitHub, always ready to take on new challenges and learn from experience.

Research Interests

SLAM / Navigation Interest in 2D/3D SLAM, Nav2, and autonomous path planning.
Simulation Building robot environments and testing programs in Gazebo, and MuJoCo.

Education

Yeungnam University *Gyeongsan, S.Korea*
B.S. IN ROBOTICS ENGINEERING *Mar. 2022 - Feb. 2026*

- Total GPA : 3.80 / 4.5
- Club : ROS Study Group - Academic Member & Project Lead

Projects

Gazebo-Based Restaurant Serving Robot Simulation **- [GitHub]** *Korea IT Academy*
SOLO PROJECT *May. 2026 - Present*

- Built a simple restaurant simulation environment where a serving robot can move and operate
- Designed the serving robot model, including its body shape and wheel structure, for basic driving
- Created a restaurant-like test space by placing tables, chairs, and a counter in the Gazebo world
- Verified basic robot operation, including model spawning, sensor checking, and simple movement control

Quadruped Robot for Fire Monitoring and Autonomous Patrol **- [GitHub]** *Yeungnam University*
PARTICIPANT *Nov. 2024 - Dec. 2025*

- Built a mapping and driving environment so that a quadruped robot can patrol outdoor areas autonomously
- Reduced unnecessary outdoor driving data to improve mapping stability
- Controlled the robot's posture and movement for stable operation on uneven ground
- Implemented a real-time alert function by detecting abnormal temperature with a thermal camera

Real-Time On-Device Hand Gesture Recognition System **- [GitHub]** *Yeungnam University*
PARTICIPANT *Mar. 2025 - Jun. 2025*

- Created a system that recognizes hand gestures from camera video and classifies custom gestures
- Extracted key hand joint positions and converted them into data for gesture recognition
- Built a custom gesture dataset using hand gesture images collected by the team
- Implemented a real-time recognition pipeline that can run quickly on mobile or PC environments

Person-Following Mobile Robot System *Yeongnam University*
PROJECT LEAD *Apr. 2024 - Sep. 2024*

- Designed the hardware and control structure to build a mobile robot that can recognize and follow a person
- Built the robot frame and wheel driving structure using aluminum profiles
- Implemented a vision function to detect a person and locate the target in the camera view
- Developed wheel driving and steering control by connecting a motor driver with a control board

Skills

Programming	C++, Python, Matlab
Robotics & DevOps	ROS, Linux, Git, Docker
Languages	Korean, English

Certificates

Jun. 2026	**Engineer Information Processing (Written Exam Passed)**, HRDK (Human Resources Dev. Service)
Mar. 2026	**ABEEK Graduate (Washington Accord)**, ABEEK Korea
Mar. 2026	**Robotics Professional (Accredited)**, ABEEK Korea
Nov. 2025	**TOEIC Speaking AL (170)**, YBM
Apr. 2022	**Computer Literacy (Grade 2)**, KCCI (Korea Chamber of Commerce and Industry)
Dec. 2018	**Type 1 Normal Driver's License**, Korean National Police Agency

Extracurricular Activity

GitHub Study Log Repository **- [GitHub]** *Github*

OPEN SOURCE CONTRIBUTOR *Apr. 2026 - Present*

- Operating a public GitHub repository to document daily learning logs
- Focusing on studying advanced robotics technologies and recording the codes continuously.
- Making study logs open to everyone to share robotics knowledge with global peers.
- Reviewing and refactoring my own source codes to build better software habits.
- Trying to learn core robot systems every day and growing as a passionate engineer.

Technical Blog **- [Blog]** *Naver Blog*

TECHNICAL BLOGGER *April. 2026 - Present*

- Writing and managing a technical blog to share new engineering knowledge
- Summarizing lectures, reviewing technical papers, and posting helpful development tips.
- Sharing what I have studied to help other developers and remember information longer.
- Contributing to the tech community and making a better society by sharing free knowledge.
- Improving technical writing skills and learning how to explain complex ideas easily.

ROS Study Group *Yeungnam University*

PROJECT LEAD & ACADEMIC ORGANIZER *Apr. 2024 - Dec. 2025*

- Leading and organizing weekly ROS2 study meetings for robotics students.
- Managing team projects and helping members solve hardware and software problems.
- Sharing study materials and robot source codes on GitHub to help the team grow.
- Studying core robotics technology like SLAM, Nav2, and simulation programs together.
- Learning how to collaborate with other software and hardware engineers effectively.